MAD Course Handout (Batch 2023-2027) - (Even Sem - 2025)

Course Handout		
Course Number: CSE3709	**Course Name:** Mobile Application Development	
Credits: 2 (1 L – 0 D – 2 P)	**Contact hours per week:** 03 hours	
Instructor-in-charge: Dr. Nikhil Kumar (Course Coordinator) Dr. Harsh Taneja Mr. Gautam Gupta	**Email Id and Contact Number:** nikhil.kumar@bmu.edu.in (9258787980) harsh.taneja@bmu.edu.in (7838797610) v-gautam.gupta@bmu.edu.in (9873198344)	

Aim of the course:

This course introduces students to programming technologies, design and development related to mobile applications. Topics include accessing device capabilities, industry standards, operating systems and programming for mobile applications using an OS Software Development Kit (SDK). Upon completion, students should be able to create basic applications for mobile devices.

Prerequisites: Java Programming Language

Course Outcomes:

At the end of the course the students should be able to do the following:

CO1: Understand the Android OS and its underlying architecture, and Android Studio.
CO2: Apply different views and layout resources for a given real-life scenario.
CO3: Design and develop Android applications and run them over real devices and Android virtual devices (AVDs).

Mobile Application Development

Topics	Number of Sessions	Course Outcomes
What is Android, Android versions and its feature set, Android Development Environment – System Requirements, Android SDK, Installing Java, Creating Android Virtual Devices (AVDs) Android Runtime - Dalvik Virtual Machine	2	CO1
Android Runtime – Core Libraries, Creating an Activity, Running the Application in the AVD, stopping a Running Application Reviewing the Layout and Resource Files	2	CO1
Intent Overview, Implicit Intents, Creating the Implicit Intent Example Project, Explicit Intents, Creating the Explicit Intent Example Application, Intents with Activities	4	CO2, CO3
Designing for Different Android Devices, Views and View Groups, Android Layout Managers, Toast, Log	6	CO2, CO3
Programming the display, Keyboard / touch, accessing camera, multimedia (Audio and Video), on device sensors	4	CO2, CO3
Sending SMS, sending email, sending notifications, Dialling a number	4	CO2, CO3
Introduction to location-based service, configuring the Android Emulator for Location Based Services, Geocoding and Map-Based Activities	3	CO2, CO3
Directory Structure and Saving Data in Files	5	CO3
Testing and deploying these applications on Emulator and on the real device	2	CO3
Total Contact Sessions:	**32**	

CO-PO Mapping

Course (Level) and Course Outcome Mapping to Program Outcomes *[Three Levels of Mapping: 1: means weak, 2: means medium and 3: means strong]*

Course (Level) Mapping PO1		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
Mapping of Course Outcomes	CO1	2	2														
	CO2	3	3	1	1	1						2	1		1		
	CO3	3	3	1	1	1					2	2	1		1		

Textbooks:

1. John Horton, Android Programming for Beginners, Third Edition, Packt Publishing, 2021
2. Neil Smyth, "Android Studio 4.1 Development Essentials - Java Edition", First Edition, Payload Media Inc., 2020

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the End-Term Project evaluation along with one assignment and two quizzes. Relative grading method defined in the academic regulations of the university will be followed to grade the students. Student must secure minimum 40% of marks after completing all the assessments in the following table to become eligible for grading.

Component	Marks	Weightage	Schedule	Remark
Quiz	2 x 20 = 40 marks	40%	One before Mid-Sem and another after Mid-Sem	There will be two quizzes in Offline mode. Each quiz will have 20 objective-type questions.
Assignment	1 x 20 = 20 marks	20%	After 2-3 weeks of Mid-Sem	Assignment would have few very small Android applications to implement.
End Term Project	Phase-1: 10 marks Phase-2: 30 marks	40%	Phase-1: A week before the Mid-Sem Phase-2: Last week of April	It will have two phases: In phase-1 (10 Marks) students will bring the Figma designs of the app, and in phase-2 (30 marks), they will show the complete running project (app), presentation, and viva-voce.

Experiential Learning: 80%

Student Responsibilities:

- Attend lectures and do labs as per instructions.
- Prepare for discussions on analysis part on lab work during defined class in every week.

Attendance Policy:

Students are expected to attend the classes regularly. Failure to attend the classes regularly and adhere to the expected attendance percentage.

Lab:

Lab work will have an analysis part and will be discussed in the defined class in every week. Based on the performance on analysis and discussion part, the marks will be given. Late submission in assignment is not allowed and any late submission will be awarded "0" marks in that assignment

Recourse examination policy: As per University policy.

Make-up policy:

No make-up work will be given for unexcused absences. The faculty needs to be informed in advance in case the student is not going to be able to submit an assignment or take any evolution component, and it is at the discretion of the faculty to sanction make up for an evaluation component.

Academic dishonesty/cheating/plagiarism:

Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.